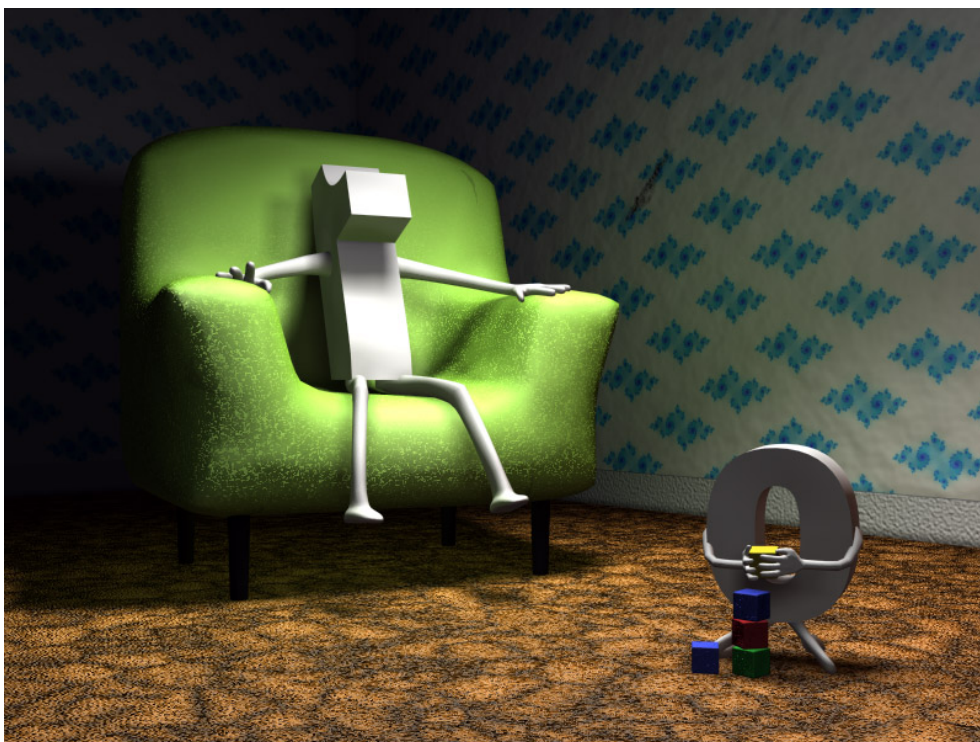


Learning how to prove Workshop Week 2

Integers



Prime Factorisation. Integer modulo n

1. **Let a and b be positive integers. If $a = a_1c$ and $b = b_1c$, where $c = \gcd(a, b)$, then $\text{lcm}(a, b) = a_1b_1c$.**

2. **Find all solutions in $\mathbb{Z}_n$ (as indicated) for each of the given equations:**

$$(a) \quad \left. \begin{array}{rcl} \bar{3}x + \bar{2}y & = & \bar{1} \\ \bar{5}x + y & = & \bar{1} \end{array} \right\} \text{ in } \mathbb{Z}_{11}, \quad (b) \quad \left. \begin{array}{rcl} \bar{3}x + \bar{2}y & = & \bar{1} \\ \bar{5}x + y & = & \bar{4} \end{array} \right\} \text{ in } \mathbb{Z}_7$$